

Jonathan Orellana

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 7 years of experience turning complex business problems into elegant, automated systems — from applying \$15M in unidentified funds daily with a purpose-built ETL pipeline, to orchestrating warehouse replenishment pipelines in Airflow and building AI agent tooling in Rust. I bring a systems-level mindset focused on scale, operational hygiene, and agentic workflows.

ENGINEERING PHILOSOPHY

Destroy Ambiguity: Define exact success/failure states before writing code to prevent technical debt.	AI-Assisted Architecture: Leverage LLMs and Diataxis to pressure-test solutions before committing to production.
Agentic Tooling: Build purpose-built MCP servers in Rust to bridge LLM reasoning and local execution.	Rigorous Refinement: Audit for security and compliance; code must fail loudly and pass silently.

EXPERIENCE

Software Engineer II The Home Depot	2024 — Present
- Engineered cloud-native backend services for retail and supply-chain platforms using Java, Spring Boot, GCP (GKE, Cloud Run), and Apache Airflow — orchestrating pipelines that process millions of inventory records daily to drive warehouse replenishment decisions across seasonal demand cycles. - Designed and maintained event-driven data workflows using Kafka, Pub/Sub, BigQuery, and SQL, enabling near-real-time operational reporting and downstream decision-making at enterprise scale. - Increased engineering velocity and production stability by refactoring legacy components, expanding automated test coverage, and hardening CI/CD pipelines — delivering safer, fully automated deployments to production.	
Systems Analyst Norfolk Southern	2021 — 2024
- Rebuilt and modernized critical internal applications (time clock, ticketing system, check inventory, security console) using a full-stack approach (ASP.Net, MS SQL, React) to enhance functionality, streamline workflows, and improve user experience across departments. - Automated daily SOX control process using RPA to reconcile inbound/outbound charges between internal systems and SAP, ensuring data integrity for KPMG audits. - Automated quarterly SOX lease accounting survey using Microsoft PowerApps and SharePoint, eliminating manual form rebuilding and streamlining data collection and distribution. - Automated permission requests and Active Directory (AD) tasks by developing PowerShell scripts, streamlining processes and reducing the time needed for user management.	
Business Analyst Norfolk Southern	2019 — 2021
Developed and deployed an automated ETL pipeline using Alteryx and Python to identify and apply over \$15 million in unidentified customer funds daily — extracting remittance data from disparate sources (PDFs, emails, CSVs), matching against invoices in DB2, and eliminating hours of manual research per analyst per day.	

- Led a \$150,000 annual efficiency improvement by leveraging Spotfire and Alteryx to streamline cash application processes, reducing manual staffing needs.
- Empowered colleagues by mentoring staff in Python coding and regular expressions (regex) to facilitate the development of an ETL process for cleaning customer remittances, improving data accuracy and efficiency.

SKILLS & TECHNOLOGIES

Languages:	Rust, Python, JavaScript, TypeScript, Java, C#, SQL
Cloud & DevOps:	GCP (GKE, Cloud Run), Docker, GitHub Actions, Azure
Backend & APIs:	Spring Boot, Node.js, ASP.NET, Django, REST APIs, MCP Protocol
Data & Automation:	Apache Airflow, Kafka, BigQuery, Alteryx, Teradata, MS SQL Server, DB2
AI & Tooling:	MCP Servers, AI Agent Harnesses, Local LLM Integration, SQLite FTS5
Frontend:	Svelte, React, HTML/CSS, Vite

EDUCATION

Bachelor of Business Administration, Finance | Kennesaw State University

2015 — 2019